
Lab Assignment # 10

Name of Student : CHATHRABOINA SAGAR
Enrollment No. : 2303A51522
Batch No. : 22

Task Description -1(Error Detection and Correction)

Use AI to analyze a Python script and correct all syntax and logical errors.

Sample Input Code:

```
def calculate_total(nums)
sum = 0
for n in nums
sum += n
return total
```

Expected Output-1:

Corrected and executable Python code with brief explanations of the identified syntax and logic errors.

Prompt:-

generate a python program to correct the code syntax and logical errors in python syntax and print the output to be corrected and executable code with docstring and comments.

Code:-

```
# Task-1
# generate a python program to correct the code syntax and logical errors in
def calculate_total(nums):
    """
    Calculate the total sum of numbers in a list.

    Args:
        nums: A list of numbers to sum.

    Returns:
        The sum of all numbers in the list.
    """
    # Avoid using 'sum' as variable name since it shadows built-in function
    total = 0
    for n in nums:
        total += n
    return total

# Test the function
if __name__ == "__main__":
    numbers = [10, 20, 30, 40, 50]
    result = calculate_total(numbers)
    print(f"Total sum: {result}")
```

Output:-

```
Total sum: 150
```

Justification:

The code had an error because sum was used as a variable name, which conflicts with Python's built-in function. So it was changed to total to fix the problem

Task Description -2(Code Style Standardization)

Sample Input Code:

```
def findSum(a,b):return a+b  
print(findSum(5,10))
```

Expected Output-2:

Well-structured, consistently formatted Python code following standard style conventions.

Prompt:-

generate a python program to comply the standard coding Style and well formatted code with proper indentation, spacing, and naming conventions to enhance code readability and maintainability.

Code:-

```
# Task-2  
# generate a python program to comply t  
def find_sum(a, b):  
    """  
    Find the sum of two numbers.  
  
    Args:  
        a: The first number.  
        b: The second number.  
  
    Returns:  
        The sum of the two numbers.  
    """  
    return a + b  
  
print(find_sum(5, 10))
```

Output:-

```
15
```

Justification:

The code is now compliant with standard coding style, including proper indentation, spacing, and naming conventions. The function name 'find_sum' is descriptive of its purpose, and the parameters 'a' and 'b' are clear. .

Task Description -3(Code Clarity Improvement)

Use AI to improve code readability without changing its functionality.

Sample Input Code:

```
def f(x,y):  
    return x-y*2  
print(f(10,3))
```

Expected Output-3:

Python code rewritten with meaningful function and variable names, proper indentation, and improved clarity.

Prompt:-

generate a python program to improve the code efficiency and performance and functionality

Code:-

```
# Task-3
# generate a python program to improve the code efficiency and performance and functionality
def calculate_adjusted_difference(base_value, multiplier):
    """
    Calculate the difference between a base value and twice a multiplier.

    Args:
        base_value: The initial value to subtract from.
        multiplier: The value to multiply by 2 before subtracting.

    Returns:
        The result of base_value minus (multiplier * 2).
    """
    return base_value - multiplier * 2

# Test the function
if __name__ == "__main__":
    result = calculate_adjusted_difference(10, 3)
    print(f"Adjusted difference: {result}")
```

Output:-

```
Adjusted difference: 4
```

Justification:

The original code had a syntax error due to using 'sum' as a variable name, which is a built-in function in Python. By renaming it to 'total', we avoid this conflict and ensure the code runs correctly. Additionally, the function now includes a .

Task Description -4(Structural Refactoring)

Use AI to refactor repetitive code into reusable functions.

Sample Input Code:

```
print("Hello Ram")
print("Hello Sita")
print("Hello Ravi")
```

Prompt:-

generate a python program using reusable functions with modular design to break down the code into smaller, reusable functions that perform specific tasks, to improve code organization and maintainability.

Code:-

```
# Task-4
# generate a python program using reusable functions
def print_greeting(name):
    print(f"Hello {name}")

print_greeting("Ram")
print_greeting("Sita")
print_greeting("Ravi")
```

Output:-

```

Hello Ram
Hello Sita
Hello Ravi

```

Justification:

The code is now organized into a reusable function 'print_greeting' that takes a name as an argument and prints a greeting message. This modular design allows for better code organization and maintainability, as the greeting logic is encapsulated within a single function that can be easily reused with different names.

Task Description -5(Efficiency Enhancement)

Use AI to optimize Python code for better performance.

Sample Input Code:

```

numbers = [ ]
for i in range(1, 500000):
    numbers.append(i * i)
print(len(numbers))

```

Expected Output-5:

Optimized Python code that achieves the same result with improved performance.

Prompt:-

generate a python program to optimize the code with improve the performance and efficiency by using appropriate data structures, algorithms, and built-in functions to optimize the code's performance and efficiency, and show the time complexity of the optimized code.

Code:-

```

# Task-5
# generate a python program to optimize the code with improve the performance
def generate_squares(n):
    """
    Generate squares of numbers from 1 to n-1 using a generator.
    Args:
        n: The upper limit (exclusive) for generating squares.
    Returns:
        A generator object that yields squared values.
    Time Complexity: O(1) for creation, O(n) for full iteration
    Space Complexity: O(1) - uses generator instead of storing all values
    """
    return (i * i for i in range(1, n))

# Optimized approach using generator
if __name__ == "__main__":
    # Generator approach - memory efficient
    squares_gen = generate_squares(500000)
    count = sum(1 for _ in squares_gen)
    print(f"Count of squares: {count}")

    # Alternative: Using list comprehension (if storage needed)
    # squares = [i * i for i in range(1, 500000)]
    # print(len(squares))

```

Output:-

Justification:-

Generators yield values lazily on-demand, avoiding storage of all n squares in memory ($O(1)$ space vs $O(n)$), ideal for large n like 500,000 to prevent memory overflow. Iteration over generator is comparably fast to lists here but scales better under memory pressure.